Higher National Diploma in Engineering
Fourth Year, First Semester Examination – 2023

EE3211 - Electrical Machines Control & Protection

Instructions for Candidate:

Answer any **Five (05)** questions
All questions carry equal marks

No. of Questions: 06
No. of Pages: 04
Time : 3 (hours)

1.

i.) Describe the operation of the rotor resistance starter with the relevant circuit diagram of starter [4 marks]

ii.) Compare the star-delta starter with the auto-transformer starter. Clearly mention their similarities and differences

[4 marks]

iii.) State two speed controlling methods used to control the speed of a series DC motor and briefly describe one of them

[2 marks]

iv.) A 240 V, shunt dc motor, drives a constant power load ($torque \propto {}^{1}/_{speed}$) and it takes 20 A from the supply and run at 2000 r.p.m. Armature resistance is 0.75 Ω and field resistance is 350 Ω. Speed of the motor is controlled by varying the supply voltage using H-bridge motor drive. It is required to run this motor with 800 r.p.m speed for some particular application;

a.) What is the required supply voltage to this application

[6 marks]

b.) Calculate the armature current drawn from the supply when running at 800 r.p.m

[2 marks]

c.) What is the field current for this application

[2 marks]

[Total 20 marks]

2.

i.) Compare the regenerative breaking and dynamic breaking of induction motor and describe their advantages and disadvantages. What is the best breaking method from these two [5 marks]

ii.) 220 V, shunt dc motor has field resistance of 550 Ω and armature resistance is 0.3 Ω. It draws 15A current when running with 1000 r.p.m under full load condition. Maximum allowable motor current is 125% full load current. Find the additional series resistance required under following breaking conditions;

a.) Plugging breaking [3 marks]

b.) Dynamic breaking [3 marks]

iii.) Describe the drawbacks of speed control of induction motors using voltage control method [5 marks]

iv.) Briefly describe the frequency control method for induction motor speed control [4 marks]

[Total 20 marks]

3.

i.) What are the factors that need to consider when selecting induction motor speed controlling method for some particular application [4 marks]

ii.) Pulse width modulation (PWM) based control is one method employed in those VFD s. Briefly explain the advantages of using PWM control in VFD s for induction motor speed control [4 marks]

iii.) Describe the rotor resistance speed control of wound rotor induction motors [4 marks]

iv.) A 4-pole three phase, delta connected, 400V, induction motor speed control of the motor is done using voltage control method. Motor run at 1460 r.p.m under full load condition. Stator impedance is negligible and rotor reactance is 3.6 times the rotor resistance. Motor load torque is constant. Find;

a.) Speed of the motor when, the supply voltage is 320V [4 marks]

b.) If we required to obtain a speed of 1200 r.p.m, what is the supply voltage [4 marks]

[Total 20 marks]

4.

i.) Synchronous motor drives a constant power load and its excitation is varied. Under excitation current, motor operate with 0.8 power factor lagging and load angle is 15° excitation current is increased by 40%, what is the new power factor of the motor. Synch reactance of the motor is 2.1 Ω .

[5

ii.) What is automatic voltage regulator (AVR) use with synchronous generator and briefly the requirement of that and how it works

[5

iii.) Briefly explain why synchronous motors are not self-starting

[3

iv.) A 75-kW, 3-φ, star-connected, 50 Hz, 450 V, cylindrical rotor synchronous motor ope rated condition with 0.85 p.f. leading. The motor efficiency excluding field and stator lo 95% and X_s= 2.65 Ω. Calculate;

a.) Armature current

[1

b.) Induced e.m.f.

[3

c.) If the excitation current is decreased by 20% while delivering the same load, what new power factor of the motor

[3

[Total 20 m

5.

i.) Briefly describe what is **hotspot temperature** of the motor and its significance in selection

[3

ii.) A motor needs to operate continuously with 9.6 kW for 2 hours, 2kW for 4 hours, 7 kW hours and 5kW for 3 hours.;

a.) Calculate the continuous equivalent power for the motor

b.) Following motor options are available to select and find the suitable motor (most economical. Assume that the price is proportional to the motor rating) for the above application;

Option 1 – 10 kW motor that has no overloading capability

Option 2 – 8 kW motor capable of operating with 125% the rated power

Option 3 – 6 kW motor capable of operating with 125% the rated power

[2+4 = 6 m

iv.) Rated power of a motor is 16 kW and thermal time constant of the motor is $\tau = 15$ minutes. Rated temperature rise of the machine is 60 ^{0}C and ambient temperature is 40 ^{0}C. Ratio between full load copper loss and iron loss is 1.4; Find the temperature of the motor when it operates at half load condition for time period of 7 minutes

[7 marks]

[Total 20 marks]

6.

i.) State **two** generator excitation methods and describe them

[4 marks]

ii.) Briefly describe the operation of electronic overload relays and compare it with bimetallic thermal overload relays

[7 marks]

iii.) Explain how the trip class of thermal overload relay is selected to provide protection for a

[5 marks]

iv.) What is the main duty of damper winding in synchronous generator

[4 marks]

[Total 20 marks]